

Kinematic Study of a Parallel-Axis Variant of the Tri-pterion Mechanism

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1.1 Introduction

Classical 3T1R (three translational and one rotational degree of freedom (DOF)) mechanisms such as Gosselin's Tri-pterion [1] and the constraint-based variants developed by Rolland and Gogu [2], [3] demonstrate that symmetry and limb decoupling can simplify mobility evaluation and kinematic analysis. In contrast, parallel-axis layouts shown by the Parallel Axis Tri-pterion streamlined the requirement for mutually orthogonal prismatic guides, allowing all linear rails to be arranged in parallel and improving assembly, alignment, and scalability. The mechanism studied in this report is inspired by several available prototype demonstrations, including a compact parallel-axis Tri-pterion built by independent developers [4], [5] and an internal research concept shared by Zaber corporation. The most notable characteristic of these prototypes is the platform's pure translational motion without rotation which produces a very clean and predictable toolhead trajectory that is uncommon in many parallel manipulators. Although these prototypes are not formally documented, they all share the same basic arrangement: two limbs input the motion, and the remaining limbs follow passively to maintain the platform's constrained motion. This property results reduced 2-DOF translational motion particularly suitable for planar and large-scale positioning tasks. Therefore, this report analyzes a simplified parallel-axis Tri-pterion, focusing on its mobility, forward kinematics, and singularity structure.

2.1 Mobility analysis

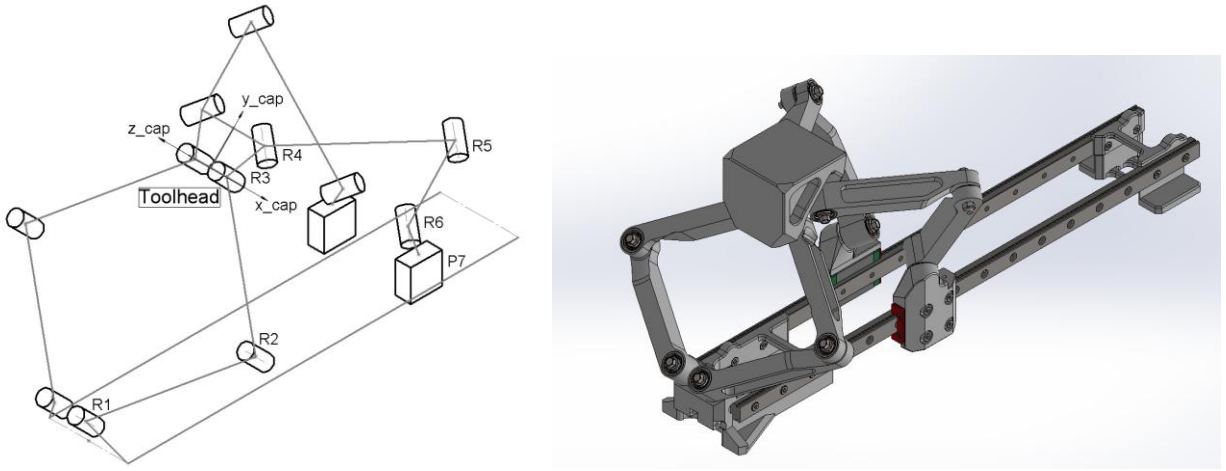


Figure 1. Structural configuration and assemble model of the 3-DOF parallel mechanism.

Applying the classical spatial Grübler's formula to evaluate DOFs,

$$DOF = 6 \times (N - 1 - J) + \sum_{i=1}^n fi, \text{ yields } DOF = 6 \times (12 - 1 - 14) + 14 = -4,$$

which is clearly inconsistent with the actual DOFs. Rolland [4] noted that, for parallel mechanisms, this formula overestimates mobility because it ignores internal parasitic motions that do not influence the end-effector. To correct evaluation of mobility, the operational DOF of the end-effector by writing the constraint equations at the platform is better to get exact outcome.

Denote that the toolhead twist by $S_i = [\omega; v]'$, and the limb constraints can be expressed as $CS_i = 0$, where C is the constraint Jacobian built from the limb constraint screws. In the front limb, the three revolute joint axes are parallel; In the rear limb, the three revolute joint axes are also parallel, but their direction is tilted by approximately 30–35° with respect to the axes of the front limb. Each limb contributes two non-collinear rotational constraints. These two non-parallel axis-planes generate three independent rotational constraints in total. Thus, these two sets of non-parallel screws span a three-dimensional subspace of rotation, and only three of them are linearly independent, causing $\text{rank}(C_\omega) = 3$, and since the toolhead has three independent rotational constraints, its instantaneous angular velocity is $\omega = 0$, $DOF_{task} = 6 - \text{rank}(C_\omega) = 6 - 3 = 3$.

Thus, this mechanism provides three pure translational degrees of freedom at the toolhead.

2.2 Forward kinematics analysis

Unlike the orthogonal Tri-pterion whose three actuated prismatic axes are aligned with the global X, Y, Z axes, the present mechanism does not possess a natural Cartesian alignment. Since the mechanism is strictly symmetric, the analysis can be reduced to only right-hand kinematic chain. The platform frame $\{c\}$ is defined at the geometric intersection of the toolhead axis and neutral plane of four branches.

For each limb i , the joints motion is represented using screw coordinates:

$${}^bT_c(q_i) = \prod_{j=0}^3 (e^{S_{jqj}})^{b_i} T_c(0) = Tp(\rho_1)T1(\theta_1)T1(\theta_2)T1(\theta_3)M_i,$$

where $q_0 = \rho_1$ represent the prismatic joint, and $q_1, q_2, q_3 = \theta_1, \theta_2, \theta_3$ are the three passive revolute joint angles of each limb.

To better analyze forward-kinematics and Jacobian, this mechanism is reduced to four planar symmetric subsystems. The two left limbs retain only their three passive revolute joints (RRR), and two right limbs sustain their actuated prismatic joint followed by three revolute joints (RRRT). As a result, the platform is constrained to its neutral plane and exhibits two translational degrees of freedom (2T).

For each joint i , the spatial twist S_i of corresponding linkage motion can be written in the standard format $S_i = [\omega; v]'$, where $\omega_i \in \mathbb{R}^3$, $v_i = -\omega_i \times q_i + h\omega_i$, with q_i is any point on the joint axis, and h is equal to zero (pure rotation). By leverage this formula and Bröker–Hunt (BH) convention, each axis can be express in the table below,

i	q_i	ω_i	v_i
1	[10, -5.5, 15.3]	[0, 0, 1]	[-5.5, -10, 0]
2	[73.64, 26.77, 73.03]	[0, 0, 1]	[26.77, -73.64, 0]
3	[10, 71.77, 118.03]	[0, 0, 1]	[71.77, -10, 0]
4	[13.47, 97.07, 131.03]	$[\sqrt{2}/2, -\sqrt{2}/2, 0]$	[92.65, 92.65, -78.16]
5	[58.99, 173.73, 118.76]	$[\sqrt{2}/2, -\sqrt{2}/2, 0]$	[83.98, 83.98, -164.56]

6	[4.528, 187.82, 41.784]	$[\sqrt{2}/2, -\sqrt{2}/2, 0]$	[29.55, 29.55, -143.082]
7	[27.8, 183.32, 25.3]	[0, 0, 0]	[0, 1, 0]

Table 1. Bröker–Hunt (BH) convention.

The home position is defined by aligning all three passive revolute joints of each limb in a mutually orthogonal arrangement, aligning with the geometric reference in Figure 1. The tool frame $\{c\}$ is attached at the measured toolhead point, and its orientation is a certain value of a fixed 45° rotation about its x-axis.

$$M = T_{home} = \begin{bmatrix} Rx(45^\circ) & [px \ py \ pz]' \\ 0 & 1 \end{bmatrix};$$

$$\text{Where } Rx(45^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}, p_{home} = \begin{bmatrix} 27.8 \\ 105.2 \\ -86.4 \end{bmatrix}$$

Hence, the forward kinematics of this mechanism is expressed through the Product of Exponentials Formula (PoE), which maps on the Special Euclidean group SE (3) and denote as,

$$e^{[S]\theta} = \begin{bmatrix} e^{[\omega i]\theta} & I\theta + (1 - \cos\theta)[\omega i] + (\theta - \sin\theta)[\omega i]^2 \\ 0 & 1 \end{bmatrix},$$

where ωi is the skew-symmetric matrix, the $e^{[\omega i]\theta}$ is identity matrix for the pure translation in P7, by substituting all exponential blocks into the PoE product can get the complete closed form expression for the pose of the tool frame $\{c\}$.

2.3 Singularity analysis

The singularity characteristics of this mechanism can be evaluated by the body Jacobian formula. The body formulation is adopted since it is set to incorporate the home configuration $M = T_{home}$, and it can express all joint twist $Bi = Ad_{M^{-1}} Si$ in the common reference frame. The instantaneous speed $Vb = [\omega b; vb]' = Jb(q) \dot{q}$,

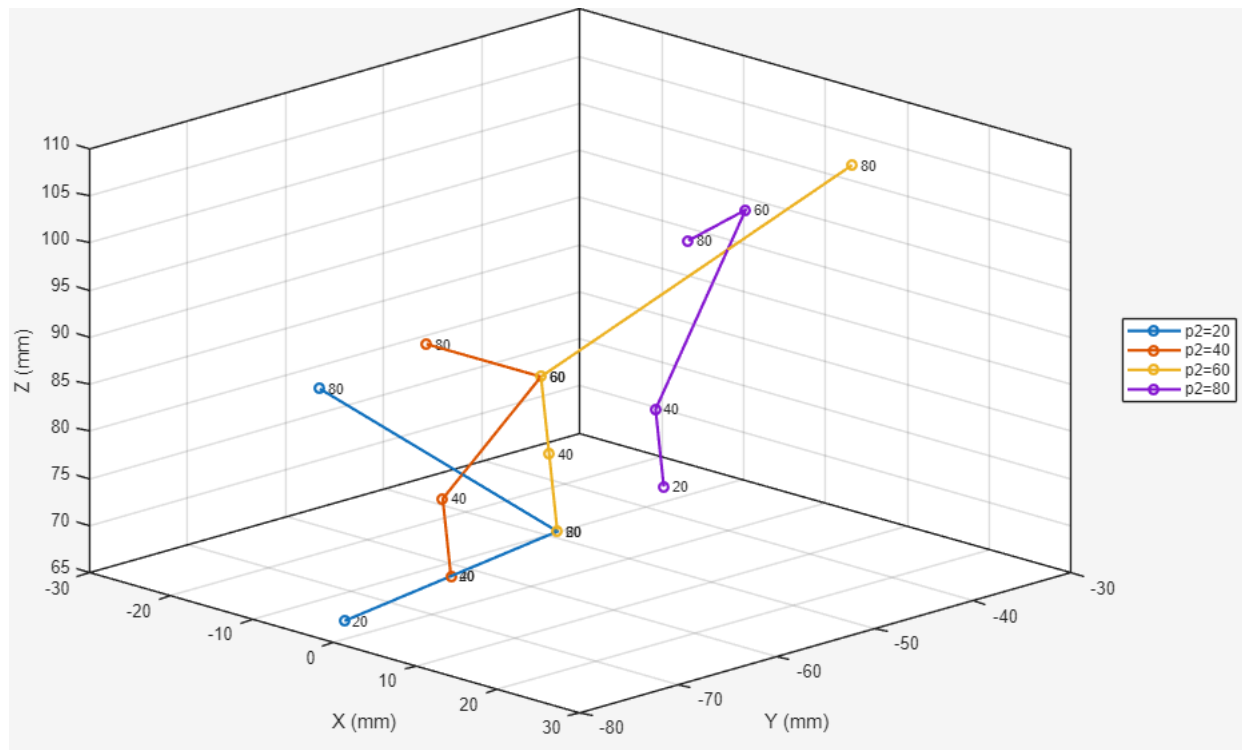
where $Jb(q) = [Ad_{e^{-B2q2} \dots e^{-B7q7}} B1, Ad_{e^{-B3q3} \dots e^{-B7q7}} B2 \dots, B7]$, $\dot{q}$ is the vector of joint positions.

Because the toolhead frame $\{c\}$ is restricted to pure translation in the neutral plain (2T), merely the in-plane parts of the end-effector velocity are relevant to the task, which can be express in a selected matrix:

$E \in R^{2 \times 6}$, $E = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$, which can express the two translational parts from the body twist. Moreover, only the two RRRT limbs provide driving prismatic joints, the task Jacobian is gained by taking according to columns of the body Jacobian. Thus, the task singularity of the toolhead happens when the rank $(J_{task}(q)) < 2$, refers to that the two actuators become linearly dependent and the platform loses one translational DOF.

2.4 Range of motion (ROM) analysis

This parallel mechanism is a closed -loop architecture and cannot get a straightforward workspace computation, which is not similar like manipulator analysis that learned in textbook. To avoid excessive computational complexity, a reduced sampling method is adopted. Therefore, the workspace analysis focuses on the reduced sampling region: $P1 \in [20, 80] \text{ mm}$, $P2 \in [20, 80]$, which aligns to the range where the assembly model remains fully stable without conflicts. Although this region is discrete and much smaller than the theoretical full ROM, it provides a reliable and non-collision condition which is sufficient to observe its translational behavior. The end-effector position are manually sampled in Solidworks by fixing one slider and evaluating the {c} frame point. The results are plotted in MATLAB.



3.1 Discussion

Although the current Jacobian analysis can be straightforward to ascertain non-singular operational region, but the interpreting of its results is the boudrary what it can convey. This suggests that closed-loop mechanism remains a more expressive and innovative way to capture these geometric constraints more specifically.

The youtube video url is here: <https://youtube.com/shorts/AcKs9vsUSMY?si=k9GrARIJ0FiKffoS>

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